

Multi-Agent Reinforcement Learning for Local Energy Market

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Introduction

As the energy transition accelerates, decentralized renewable generation becomes more widespread, electrical distribution networks are evolving into complex, dynamic systems. These active networks involve a diverse set of stakeholders - such as producers, consumers, and system operators - who must share a common infrastructure. Managing these shared assets poses significant challenges, particularly when the system is under stress and the individual interests of actors may conflict with the collective good [1].

A key issue arises in situations of congestion, where the optimal strategy for an individual agent - seeking to maximize its own benefit - can be detrimental to the overall stability of the system. This phenomenon, known as the tragedy of the commons, highlights the difficulty of achieving coordinated behavior when individual decisions have only marginal impact but collective outcomes are critical.

To address these challenges, it becomes essential to design dynamic, understandable, and effective rules for network usage and pricing, especially under congested conditions. Reinforcement learning, and more specifically multi-agent reinforcement learning (MARL), offers a promising framework to model and analyze these decentralized decision-making processes. By enabling agents to learn adaptive strategies in response to both system dynamics and the behavior of others, MARL can support the development of robust mechanisms for regulating consumption, managing congestion, and aligning individual actions with system-wide objectives [2].

This work focuses on exploring these ideas through the simulation of a local electricity market, where a community of agents interacts with a constrained distribution network. Through iterative learning and price-based coordination, the aim is to better understand how decentralized intelligence can contribute to the efficient and fair operation of future energy systems.

1 Agent-Based Modeling and ADMM Resolution

1.1 From Centralized to Agent-Based Market Design

Traditional electricity market models often rely on a centralized or single-agent formulation, where a central planner optimizes the behavior of the entire system based on complete information. In this setup, individual preferences, constraints, and strategies of participants are aggregated into a single optimization problem. While this approach offers simplicity and tractability, it fails to capture the decentralized nature of modern energy systems, where each participant (consumer, producer, or prosumer) acts independently with limited information and its own objectives.

In contrast, a **multi-agent model** explicitly represents each actor as an autonomous agent with its own decision-making process. These agents exchange information to reach an agreement before initiating energy exchanges. In local energy markets, this enables the simulation of heterogeneous behaviors and decentralized coordination. The framework used here leverages the **Spade** platform, a Python-based library for simulating multi-agent systems, to define and manage agent interactions. In addition, the **Peak** library [3], developed at GECAD and based on **Spade**, is used to simplify the implementation of agent interactions and data exchanges. Agents exchange messages through GECAD's XMPP server¹, whose domain name is *xmpp.gecad.isep.ipp.pt*. Each agent is equipped with a local objective function and communicates within the system to exchange the necessary information.

¹Communication server using Extensible Messaging and Presence Protocol to enable real-time message exchange.

1.2 Distributed Optimization with ADMM

To ensure consistent and coordinated decision-making among agents - especially when they are interconnected through shared constraints like network capacities - an appropriate resolution method is required. Here, the **Alternating Direction Method of Multipliers (ADMM)** is adopted. ADMM is particularly well-suited for distributed optimization problems because it decomposes a global problem into smaller sub-problems, which can be solved independently by each agent, followed by a coordination step to enforce consistency [4].

ADMM operates by introducing auxiliary variables and dual variables to decouple the problem. Each agent solves its local problem independently, updates its local variables, and exchanges information with other agents and a central coordinator. The coordinator then updates the shared variables to enforce the global constraints. For instance, it ensures the anti-symmetry of power exchanges (i.e., if agent i sends power to agent j , then agent j receives the opposite amount) and the symmetry of the associated prices, meaning both agents agree on a common price for each exchange. This iterative process continues until convergence is reached.

2 Notations

To address this constrained optimization problem, a formal mathematical framework is established, taking into account both the physical laws of the electrical network and the dynamics of supply and demand. This framework is based on a cost function associated with each agent, which is to be minimized [5].

2.1 Mathematical Symbols and Domain Sets

The notations used throughout the model are given below.

- Λ : Matrix of transaction prices λ_{nm} from agent n to agent m
- $\mathbf{T}$: Matrix of exchanged power t_{nm} from agent n to agent m
- Γ : Matrix of network fees γ_{nm} for exchanges from agent n to agent m (price signal)
- p_n : Total exchanged power by agent n
- $\underline{p}_n/\overline{p}_n$: Lower/upper bounds of the power exchanged by agent n
- $\underline{t}_n/\overline{t}_n$: Lower/upper bounds on the power that agent n can exchange
- Ω : Peer-to-peer market community composed of N_Ω agents
- Ω_c : Set of consumers (subset of Ω)
- Ω_g : Set of producers (subset of Ω)
- Ω_p : Set of prosumers (subset of Ω)
- ω_n : Set of trading partners of agent n
- f_n : Cost function of agent n , defined by the coefficients a_n and b_n

2.2 Formalization of the Optimization Problem

These notations lead to the following relationships:

$$\mathbf{T} = -\mathbf{T}^\top \quad (1)$$

$$\Lambda = \Lambda^\top \quad (2)$$

$$\Gamma = \Gamma^\top \quad (3)$$

$$p_n = \sum_{m \in \omega_n} t_{nm}, \quad n \in \Omega \quad (4)$$

$$\underline{p}_n \leq p_n \leq \overline{p}_n, \quad n \in \Omega \quad (5)$$

$$p_n \geq 0, \quad n \in \Omega_g \quad (6)$$

$$p_n \leq 0, \quad n \in \Omega_c \quad (7)$$

$$\underline{t}_n \leq t_{nm} \leq \overline{t}_n, \quad n \in \Omega, \quad m \in \omega_n \quad (8)$$

In this setting, each agent is defined by a cost function, typically modeled as a quadratic function of the form:

$$f_n(p_n) = a_n p_n^2 + b_n p_n$$

where p_n represents the power consumption or production decision of agent n . These functions are convex as long as $a_n > 0$, which guarantees the existence of a unique global minimum. The overall objective of the system is to minimize the sum of all agents' cost functions, taking into account the price signals, while satisfying global constraints such as supply-demand balance and line capacity limits:

$$f = \sum_{n \in \Omega} f_n(p_n) + \sum_{m \in \omega_n} \gamma_{nm} t_{nm} \quad (9)$$

In a decentralized electricity market, agents interact locally, but their decisions are globally coupled through constraints. In particular, constraint (1) ensures the reciprocity of trades between agents, i.e., $t_{nm} = -t_{mn}$.

To enable distributed optimization, the reciprocity constraints are relaxed, where each agent n maintains local variables t_{nm} representing trades with its neighbors $m \in \omega_n$. The agreement between agents is enforced through the constraint:

$$t_{nm} + t_{mn} = 0, \quad \forall (n, m)$$

These constraints are incorporated into the objective using an *Augmented Lagrangian* formulation. Specifically, dual variables λ_{nm} are used for each consensus constraint, and a quadratic penalty term with parameter $\rho > 0$ is introduced [6]:

$$\begin{aligned} \mathcal{L}_\rho(\mathbf{p}, \boldsymbol{\lambda}) = & \sum_{n \in \Omega} f_n(p_n) + \sum_{m \in \omega_n} \gamma_{nm} t_{nm} \quad (10) \\ & + \sum_{(n,m) \in \Omega \times \omega_n} g_{nm} \end{aligned}$$

With,

$$g_{nm}(t_{nm}, t_{mn}) = \lambda_{nm}(t_{nm} + t_{mn}) + \frac{\rho}{2}(t_{nm} + t_{mn})^2$$

3 Simulated Use Case: Multi-Agent Coordination in Local Energy Exchange

3.1 System Architecture and Agent Description

In this simulation, a simplified local energy market is modeled (Figure 2) involving three active agents: a **Producer**, a **Consumer**, and a **DSO Agent** (Distribution System Operator), along with a **Global Coordination Agent** (Community Manager). Each active agent is responsible for performing local calculations to determine optimal energy exchanges based on its current state, preferences, and constraints. The community manager agent, by contrast, does not participate in energy exchanges but serves as a central coordinator to facilitate consensus and global consistency.

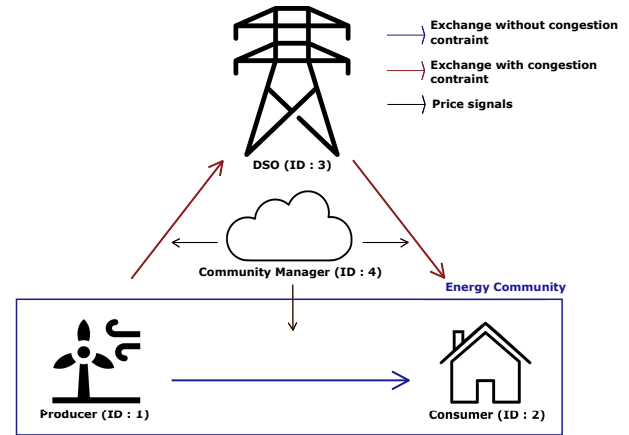


Figure 2: *Simplified Local Energy Market under Study*

Agent Roles

- **Producer** (Neighbour 1): Generates energy and seeks to distribute it to the consumer or inject it into the grid.
- **Consumer** (Neighbour 2): Demands energy and can source it from either the producer or the grid.

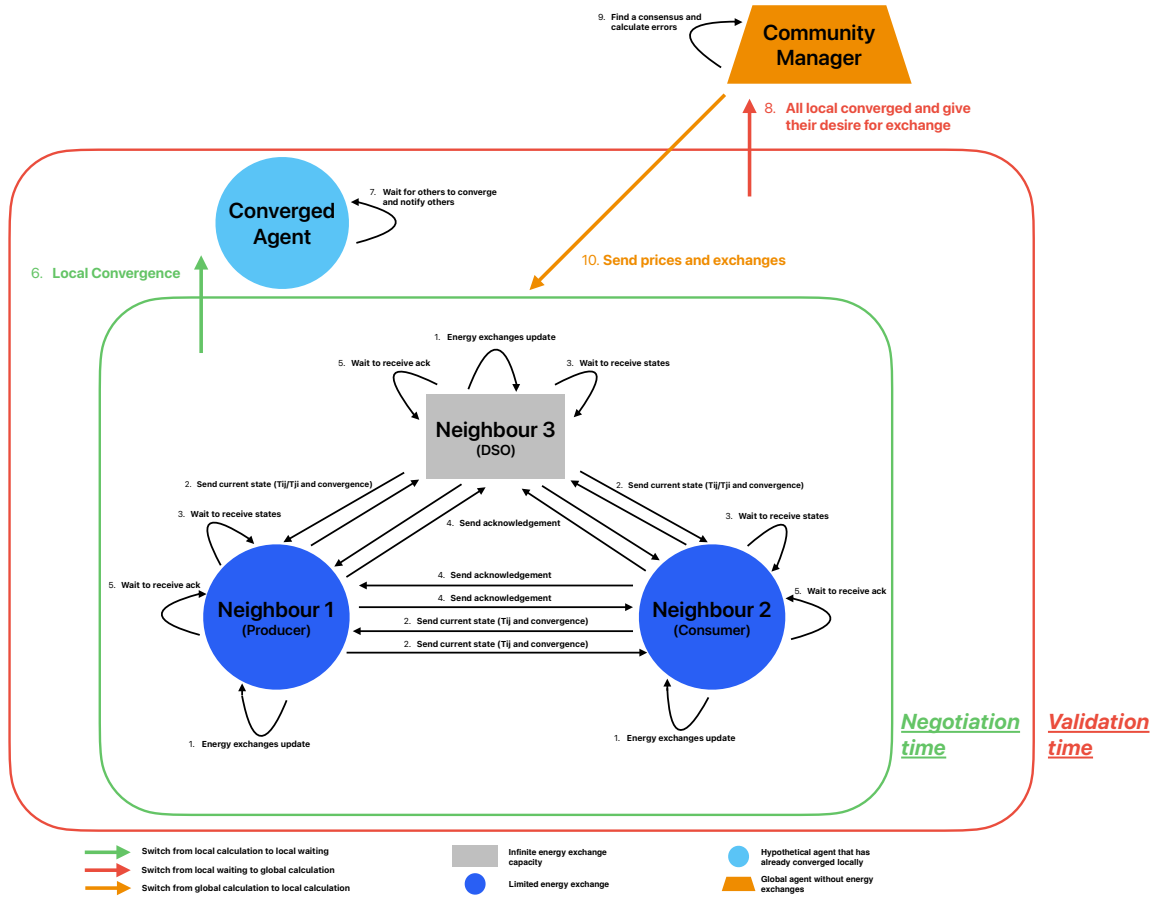


Figure 1: *Communication Process of Multi-agent Energy Market*

- **DSO Agent (Neighbour 3):** Represents the traditional power grid. As a prosumer, it can provide or absorb an infinite amount of energy.
- **Community Manager Agent:** Oversees the consensus process. It aggregates the locally converged states from the other agents and determines globally consistent prices and allocations.

3.2 Communication and Coordination Process

In a distributed multi-agent optimization process, agents do not converge simultaneously. The number of iterations required for local convergence varies due to differences in local problem structures and heterogeneous computational times. Therefore, it is essential to design a com-

munication protocol that allows asynchronous local convergence while maintaining global synchronization.

The coordination mechanism alternates between local and global computation phases and is summarized in Figure 1. The following steps describe the complete workflow:

- Each agent performs local computations to update its desired energy exchanges (T_{ij}, T_{ji}), based on local objectives and neighbor information.
- Upon reaching local convergence - typically based on residual thresholds - an agent switches from *Local Calculation* to *Local Waiting*. It sends its current state, including proposed exchange values and a convergence flag, to other agents.
- The agent then waits to receive the current

states of other agents. During this synchronization phase, it does not continue local updates.

- Once each agent has received the state updates from its peers, it sends back acknowledgements (ACKs) to confirm receipt. This ensures that no agent proceeds with incomplete or inconsistent information.
- After all agents have locally converged and the acknowledgements are received, the global agent collects the complete set of intended exchanges.
- The global agent aggregates these values, resolves inconsistencies (e.g., non-matching $\mathbf{T}_{ij}$ and $\mathbf{T}_{ji}$) using a consensus algorithm, and computes any necessary corrections in order to obtain $\mathbf{T} = -\mathbf{T}^\top$.
- It then updates shared variables such as prices and validated exchange values, which it broadcasts to all agents.
- Upon receiving this information, agents transition from *Global Calculation* back to *Local Calculation*, resuming their iterative updates with the new global signals.

This structured alternation between local autonomy and centralized coordination ensures convergence to a consistent global solution. The explicit use of acknowledgements and waiting phases prevents race conditions and ensures that all agents are aligned before proceeding to the next iteration. This approach is especially robust in asynchronous environments where local computational delays and network latencies vary. However, this implementation tends to require significantly more time to converge. The addition of a global step enforcing consensus among agents mitigates this issue and accelerating convergence [7]. A formal proof of convergence is provided in Appendix.

4 Reinforcement Learning for Congestion Management

In order to prevent congestion on the physical connection between agents and the main electrical grid, a price-based regulation mechanism is introduced. Specifically, a global agent is trained using SAC reinforcement learning to dynamically influence local agent behavior through price signals. These signals, sent at each global step, aim to guide the agents toward energy exchange decisions that alleviate stress on the grid interface.

The learning process is driven by a reward function (Figure 3) specifically designed to penalize grid congestion, thereby encouraging the emergence of more balanced and efficient exchange patterns. At each global iteration, the trained agent computes and broadcasts a set of price signals - represented by the variable $\mathbf{\Gamma}$ - to all local agents. These signals are then integrated into their decision-making processes.

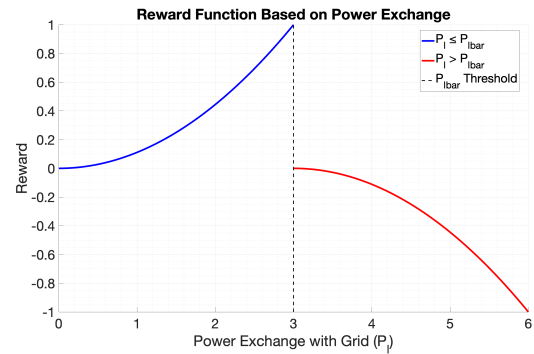


Figure 3: *Reward Function Profile for Reinforcement Learning*

Two methods have been explored to apply these price signals. The first consists in adjusting the prices of all energy exchanges, including peer-to-peer trades. The second focuses only on exchanges with the electrical grid, where congestion is most critical.

4.1 Dynamic Price Signals Across All Energy Trades

In this approach, the reinforcement learning (RL) agent generates dynamic price signals Γ that affect all energy exchanges, including both peer-to-peer (P2P) trades and transactions with the main grid. The intent is to indirectly manage congestion by modifying the local incentives of each agent, regardless of the direction or destination of their energy flow.

By incorporating the signal into every transaction, local agents are encouraged to collectively adjust their power injections or withdrawals in a way that balances the system. This broader application of Γ often results in smoother global behavior but may dilute the focus on alleviating congestion at the grid interface specifically.

To assess the impact of this strategy, we compare the resulting power flow on the congested line with and without the RL-generated price signal. As shown in Figure 9, the price signals lead to a noticeable reduction in power values on the constrained line, indicating a successful mitigation of congestion risks.

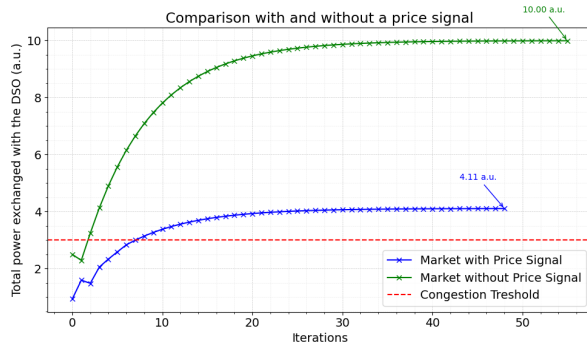


Figure 4: *Impact of Price Signals on Grid Power Exchange*

Although the learning-based controller enables congestion management in the market by adjusting price signals, it was observed that the congestion limit is still violated in some instances. This phenomenon can be explained by analyzing the shape of the reward function (Figure 3).

Initially, the agent explores actions leading to

higher rewards, quickly reaching the region near the maximum reward. However, as the iterations progress and the system converges, the reward sharply decreases due to the penalization applied when the power exchange exceeds the threshold P_l . This is a direct consequence of the reward function's structure: while it encourages increasing power exchange up to the limit P_l , it only gradually penalizes violations beyond this point. Moreover, because the ADMM procedure adjusts variables through successive local updates, early choices made during the first iterations significantly influence the trajectory toward the final solution, making later corrections less effective in fully eliminating constraint violations.

Consequently, the learned policy may tolerate slight overstepping of the congestion limit. Initially, the agent maximizes its reward by increasing exchanges with the DSO. Although penalties are applied beyond the threshold, the influence of early price signal choices on the convergence dynamics can lead to a final policy that allows minor violations. This trade-off between maximizing early rewards and the lasting impact of these choices on convergence explains the small but persistent breaches observed in the price-driven market.

4.2 Improving Congestion Compliance Through Reward Function Redesign

While the current approach reduces congestion via dynamic price signals, it does not fully ensure strict adherence to capacity limits, due to the structure of the reward function and the convergence dynamics of the learning process. To better capture this effect and mitigate its impact, new reward functions will be explored with the aim of improving congestion compliance in the system.

First, the same reward function as previously is retained, but it is shifted to anticipate the power exchanged with the DSO. This means that the penalty is applied earlier, before reaching the

constraint limit. A smooth quadratic penalty discourages getting too close to the boundary, while a strong negative reward is applied when the limit is exceeded. This anticipatory design promotes proactive constraint compliance, reduces the impact of early actions on convergence, and helps prevent violations. A regularization term also penalizes abrupt changes in the coordination variable γ , encouraging smoother temporal evolution. The shape of the reward curve remains the same, but is shifted to the left to reflect this anticipation.

While the results show slightly improved compliance, the performance gains are not significant. This suggests that the discontinuity in the reward function at the constraint threshold may hinder the learning process.

Consequently the second formulation, denoted as the *Gaussian reward* presented in Figure 5, offers a continuous incentive landscape centered around a target operating point $P_{\text{opt}} = 0.8 \cdot P_l^{\text{max}}$, with a standard deviation $\sigma = 0.2 \cdot P_l^{\text{max}}$.

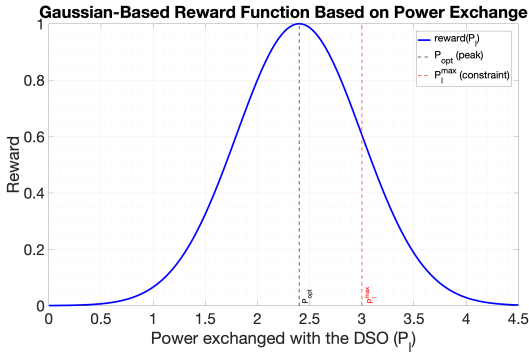


Figure 5: *Continuous Reward Function Used for Reinforcement Learning*

This configuration softly guides the system toward an optimal operating region, reducing the likelihood of extreme power exchanges and promoting overall grid stability. Similar to the previous case, a penalty on the variation of γ is applied to limit abrupt strategy shifts. The results show a significant improvement in performance. As illustrated in Figure 6, the power exchanged with the DSO in the price-signal-based market scenario is flattened from the very beginning. This early regulation enables the sys-

tem to converge much closer to the constraint, while maintaining smooth and stable dynamics throughout the horizon. The anticipatory nature of the price signals allows agents to adjust their behavior gradually and in coordination, resulting in better congestion management and more efficient use of network capacity.

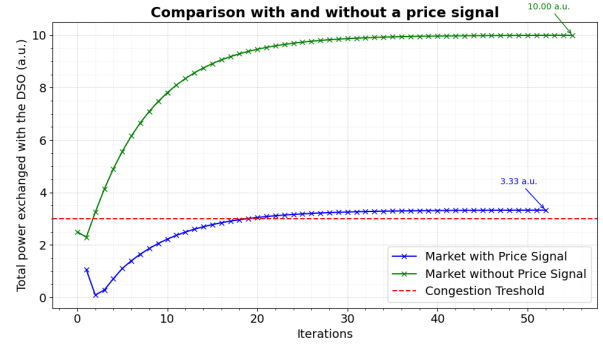


Figure 6: *Effect of Price Signals on Grid Power Exchange Using a Gaussian Reward Function*

Finally, changing the reward function from discontinuous to continuous (Gaussian) reduced the power exchanged with the grid and the congestion by approximately 23.4% compared to the initial setting. As illustrated in Figure 7, the continuous reward consistently increased throughout training, demonstrating that this approach enables more efficient and stable learning compared to the initial one, with the agent effectively adapting its behavior to maximize the designed objective.

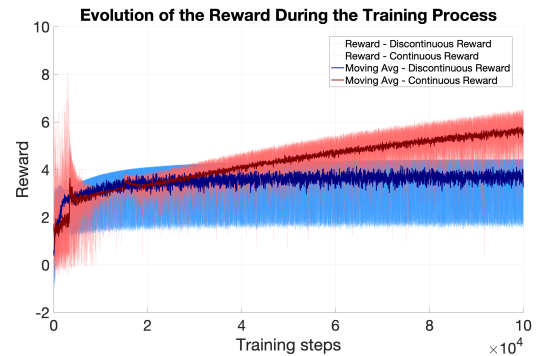


Figure 7: *Comparison of Continuous and Discontinuous Reward Functions on Learning*

In the ideal case, both reward functions lead

to the same theoretical maximum, which is approximately 55, further confirming that the performance gains are due to improved learning dynamics rather than a change in the objective itself.

The agent progressively learned to operate in a more coordinated and anticipative manner, reducing the occurrence of extreme power exchanges and avoiding overloading situations. The training process thus confirms the capacity of the proposed reward shaping and penalty design to drive the system toward grid-friendly behaviors while maintaining decentralized decision-making.

5 Learning-Based Control in Multi-Agent Energy Markets

In the next step, the objective was to integrate the learning-based congestion management model into the multi-agent framework in order to reproduce interactions that more closely resemble real-world agent behavior. Since the evolution of the interactions and the convergence point are similar between the simple market and the multi-agent setting, the trained learning-based model from the simpler configuration was chosen for deployment. This approach significantly reduces computational effort while preserving the accuracy of the results.

5.1 Consistency of Results on Congestion Between Traditional and Multi-Agent Markets

To ensure the validity of this approach, it was essential to verify that both the traditional centralized market formulation and the multi-agent market formulation model the same underlying optimization problem. Consequently, the solutions obtained through the ADMM-based resolution in both cases must converge to the same

optimal point. As illustrated in Figure 8, this consistency check confirmed that the two frameworks reach identical solutions when using the same stopping criterion. Interestingly, the multi-agent market achieved convergence in fewer iterations compared to the traditional one. This improvement is due to the presence of a community manager enforcing additional consensus among agents, which accelerates the alignment of local decisions with the global optimum.

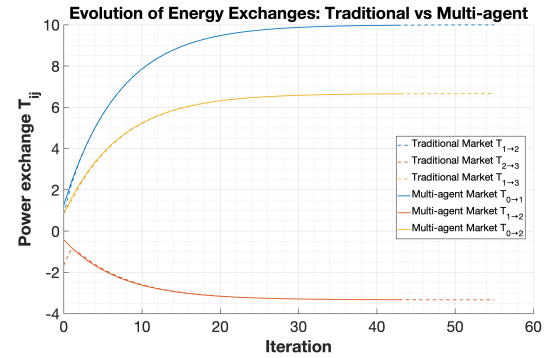


Figure 8: *Convergence Comparison between Traditional and Multi-agent Market*

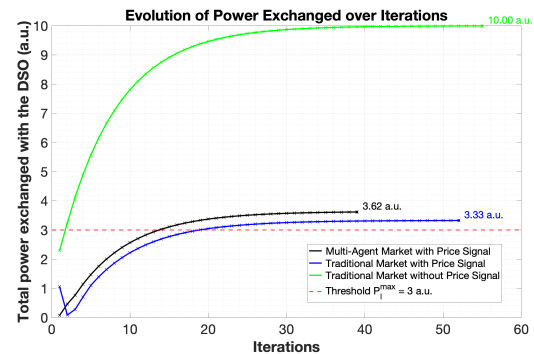


Figure 9: *Impact of Price Signals on Grid Power Exchange*

Then, congestion management within the multi-agent market becomes feasible through the use of the learning-based controller trained on the traditional market. However, because the agents' observations in the multi-agent setup are not strictly identical to those in the centralized configuration, the actions taken by the learning-based controller are slightly adapted to the new context. This adjustment leads to a slightly

different convergence behavior when the market is driven by price signals, as illustrated in Figure 9. Finally, Overall, reinforcement learning-based price signaling reduces capacity violations by nearly 90%, drastically improving adherence to network (overshoot reduced from +233% to +11%).

5.2 Realistic Congestion Management in a Dynamic Agent Environment

To enhance realism, the system models variability in agent cost functions, reflecting the nuanced decision-making of different market participants. In real-world settings, prosumers and consumers exhibit a wide variety of needs and behaviors, which are typically encoded through heterogeneous cost functions - capturing differences in preferences, consumption patterns, or flexibility levels.

The learning agent must now operate in an environment where agents no longer respond uniformly to price variations. It opens the door to more nuanced market dynamics, where the same price signal can trigger divergent reactions depending on the agent's internal model.

Incorporating such diversity not only enhances the realism of the simulation, but also tests the robustness and generalization capability of the learning agent. Reward shaping must now account for a broader range of individual agent responses, and policies must adapt to evolving configurations of agent populations. This step is essential for scaling peer-to-peer market coordination mechanisms to more complex and realistic settings, paving the way toward adaptive, resilient, and fair grid-aware pricing strategies.

To assess the real benefit of the learning process, the performance of the trained agent is compared with that of an untrained baseline in the same dynamic environment. This comparison helps evaluate not only the efficiency of the learned policy, but also its capacity to cope with unexpected behaviors and structural variability

- key indicators of robustness in real-world applications.

Figure 10 shows the convergence of the power exchanged with the DSO for an RL-based price setter. Similar experiments conducted under both fixed and variable agent cost functions yielded nearly identical convergence patterns, consistent with ADMM's properties in convex settings - where a given price trajectory leads to a unique primal-dual equilibrium regardless of initial conditions [7]. The results highlight: low variance across simulated markets at convergence, stable convergence after about 20 iterations, and consistent final performance despite different demand profiles, confirming that the learned pricing mechanism remains effective and robust across heterogeneous market configurations.

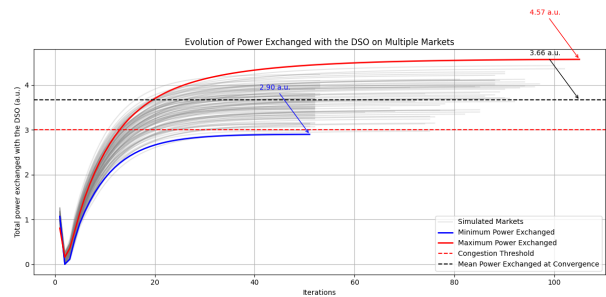


Figure 10: *DSO Power Exchange Convergence with RL Trained in a Fixed Market*

However, the convergence results are not always fully satisfactory in terms of strictly enforcing the congestion constraint, as performance strongly depends on the agents' demand profiles. A potential way to improve compliance would be to employ more extreme price signals, thereby forcing stronger adjustments in agent behavior; yet, such an approach raises concerns about the stability and convergence of the overall ADMM-based coordination process.

Conclusion and Outlook

This work explored the integration of multi-agent reinforcement learning (MARL) techniques into the control of a local energy market operating under network constraints. Starting from a centralized formulation, we designed a distributed optimization framework based on the Alternating Direction Method of Multipliers (ADMM), enabling autonomous agents - producer, consumer, and DSO - to negotiate power exchanges through iterative local computations and global synchronization.

To ensure consistency and robustness, the underlying optimization problem was with symmetric constraints and convex cost functions, and implemented a structured coordination mechanism between agents. The effectiveness of this agent-based architecture was validated by comparing its convergence behavior with a centralized resolution, showing not only equivalence of solutions but also improved convergence rates due to the explicit enforcement of consensus.

A reinforcement learning-based controller was then trained to influence agent decisions through dynamic price signals. These signals aimed to mitigate grid congestion, particularly at the interface with the main electrical network. Various reward function designs were tested, highlighting the trade-off between responsiveness and strict compliance with line capacity limits. The approach successfully guided the system toward more balanced exchange profiles while maintaining computational tractability.

Although initial results demonstrated the potential of end-to-end learning through full price signal control, a more realistic market influence strategy was subsequently explored by restricting the agent's action space to price signals related only to exchanges with the DSO. This refinement reflects practical limitations of real-world grid operators, who typically lack authority over purely peer-to-peer transactions. The key challenge of this approach was to overcome the reduced leverage of the learning agent, which required more careful reward shaping and implementation to ensure effective policy learning.

Despite the narrower scope of influence, this strategy preserved the ability to steer market behavior in a grid-supportive direction, laying the groundwork for more modular and deployable learning-based control mechanisms in distributed energy systems.

Looking forward, this work opens two promising avenues for extending the proposed framework toward more realistic and operationally relevant applications. Both directions have been preliminarily explored, but the limited time available prevented the derivation of fully interpretable or conclusive results. Nevertheless, it is anticipated that future research projects or internships will provide the necessary framework and time to successfully carry out these investigations by building upon the present work.

The first consists in managing grid congestion solely through price signals exchanged between the DSO and other market participants. By restricting the learning agent's influence to this subset of interactions, the model more accurately reflects regulatory and infrastructural constraints, while testing the agent's ability to induce grid-friendly behaviors in a decentralized market setting.

The second direction involves the integration of flexibility in the form of a controllable battery. Operated by a dedicated reinforcement learning agent, the battery would learn to respond to DSO-to-agent price signals in a way that minimizes the average cost of electricity exchanges over a given time horizon. This approach introduces a new layer of optimization, where storage is leveraged to smooth consumption patterns and arbitrage price fluctuations.

Together, these developments pave the way for a multi-agent learning architecture in which one agent manages price-based congestion signals and another coordinates battery operation. Designing effective interactions between these agents to jointly alleviate congestion and reduce energy costs represents both a technical challenge and a key opportunity for advancing intelligent control in peer-to-peer energy systems.

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Appendix

All the algorithms developed during the internship and used to obtain the results presented in this report are available on the following GitHub in .zip format: https://github.com/baptrt/GECAD_Internship

ADMM Algorithm Structure

While $(r, s) > \epsilon_g$ and $k < k_{\max}$

While $(r_l, s_l) > \epsilon_l$ and $j < j_{\max}$

Local Problem (for each agent):

$$B_{t2} \leftarrow t_i^j - \bar{t} + \tilde{p}^j - \mu^j$$

$$T^{j+1} \leftarrow \frac{\rho \cdot B_{t1} + \rho_l \cdot B_{t2} - \gamma_{ni}}{\rho + \rho_l}$$

$$T^{j+1} \leftarrow \max(\min(T^{j+1}, \text{Ub}), \text{Lb})$$

$$\bar{T} \leftarrow \text{mean}(T^{j+1})$$

$$B_{p1} \leftarrow \mu^j + \bar{t}^{j+1}$$

$$\tilde{p} \leftarrow \frac{M_n \cdot \rho_l \cdot B_{p1} - M_n \cdot b_n}{M_n \cdot \rho_l + M_n^2 \cdot a_n}$$

$$\tilde{p} \leftarrow \max(\min(\tilde{p}, P_n/M_n), p_n/M_n)$$

$$\mu \leftarrow \mu^j + \bar{t}^{j+1} - \tilde{p}^{j+1}$$

If $(j \% \text{Step}_l == 0)$

$$(s_l^{k+1}, r_l^{k+1}) \leftarrow (|t_i^{j+1} - t_i^j|, |\bar{T} - \tilde{p}|)$$

End If

End While

Global Problem:

$$\Lambda \leftarrow \Lambda + \frac{\rho}{2}(t_{nm}^k + t_{mn}^k)$$

$$B_{t1} \leftarrow 0.5(t_{ni}^k - t_{in}^k) - \frac{\lambda_{ni}^k}{\rho}$$

$$p_n^{k+1} \leftarrow \bar{T} \cdot M_n$$

If $(k \% \text{Step}_g == 0)$

$$(r^k, s^k) \leftarrow (|T - T^l|, |T^{k+1} - T^k|)$$

End If

End While

Here, M_n denotes the number of neighbors of agent n .

This algorithm makes use of ρ and ρ_l , which allow control over the convergence speed inde-

pendently for global and local updates. The variables ϵ_l and ϵ_g represent the maximum tolerable errors at the local and global levels, respectively. Finally, to ensure algorithm termination, hard stopping criteria are defined: $k_{\max}$ and $j_{\max}$, which correspond to the maximum number of global and local iterations.

Market Structure

- Producer (Agent 1) :

→ Cost function: $f_n = X^2 - 10X$;

→ Limits of the agent's power :

$$\underline{p}_n = 0 \text{ et } \overline{p}_n = 60;$$

→ Limits on agent power exchanges :

$$\underline{t}_n = \underline{p}_n \text{ et } \overline{t}_n = \overline{p}_n.$$

- Consumer (Agent 2) :

→ Cost function: $f_n = X^2 + 20X$;

→ Limits of the agent's power :

$$\underline{p}_n = -30 \text{ et } \overline{p}_n = 0;$$

→ Limits on agent power exchanges :

$$\underline{t}_n = \underline{p}_n \text{ et } \overline{t}_n = \overline{p}_n.$$

- DSO (Last Agent) :

→ Cost function: $f_n = X^2 + 10X$;

→ Limits of the agent's power :

$$\underline{p}_n = -\infty \text{ et } \overline{p}_n = \infty;$$

→ Limits on agent power exchanges :

$$\underline{t}_n = \underline{p}_n \text{ et } \overline{t}_n = \overline{p}_n.$$

With regard to the convergence parameters of the method, the following values were selected:

- $\rho = 10$;

- $\rho_l = 1$;

- $\epsilon_l = 10^{-3}$;

- $\epsilon_g = 10^{-3}$;

The other simulation variables are initialised to 0.

Proof of Invariance of the Convergence Point under Symmetric Averaging

In this section, we show that in the context of solving the distributed sharing problem (10) using ADMM, replacing the variable t_{mn}^k with the symmetric average $\bar{t}_{mn}^k := \frac{1}{2}(t_{mn}^k + t_{nm}^k)$ in the expression of the term $g_{nm}(t_{nm})$ does not alter the convergence point of the algorithm. Furthermore, this substitution often improves convergence speed and stability.

Using the notations introduced in equation (10), the resolution of the problem leads to the following function:

$$g_{nm}(t_{nm}) = \lambda_{nm}^k \left(t_{nm} + \frac{t_{mn}^k - t_{nm}^k}{2} \right) + \frac{\rho}{2} \left(t_{nm} + \frac{t_{mn}^k - t_{nm}^k}{2} \right)^2 \quad (11)$$

Here, g_{nm} is strictly convex, and so is $\mathcal{L}_\rho$, which ensures the existence and uniqueness of a solution to the problem.

Then, the ADMM algorithm converges to a solution satisfying the following conditions [8]:

$$\begin{cases} \frac{\partial \mathcal{L}_\rho}{\partial \lambda_{nm}} = t_{nm}^* + t_{mn}^* = 0 \\ \frac{\partial \mathcal{L}_\rho}{\partial t_{nm}} = \nabla f_n \left(\sum_{m \in \omega_n} t_{nm}^* \right) + \lambda_{nm}^* = 0 \end{cases}$$

At convergence, since $t_{nm}^* = -t_{mn}^*$, we have:

$$\bar{t}_{mn}^* = \frac{1}{2}(t_{mn}^* - t_{nm}^*) = t_{mn}^*$$

Therefore, the substitution $\bar{t}_{mn}^k$ is exactly consistent with the optimal solution. The minimizer of the augmented Lagrangian remains unchanged, and hence the convergence point is preserved.

Finally, replacing t_{mn}^k with its symmetric average reduces the directional asymmetry between agents at each iteration. This leads to:

- smoother dual variable updates,
- reduced oscillations between iterations,
- improved empirical convergence rates.

This aligns with theoretical results from the distributed ADMM literature [9]. Antisymmetric averaging often improves the effective condition number of the primal-dual updates.